Bayside Property Analysis - Assignment 3

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AD 571

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Question 1.

Note: this section was filtered to remove outliers and 0's

The restraints are shown below. This is based on previous use with the data.

0<SALE PRICE <= $2,500,000

0<GROSS SQUARE FEET < 5,000

a. 6,126
b. Mean Sale Price Post 2009: $840,946.70
   Mean Square Footage Post 2009: 1,841.20
   Mean Sale Price Pre 2009: $624,685.90
   Mean Square Footage Pre 2009: 1,959.27
   Difference in Price Mean: $216,260.80
      i. Sale Price post 2009 is $216,260.80 higher than pre 2009.
   Difference in Square Footage Mean: -118.08
      Square Footage is 118.08 sqft higher pre 2009 than post 2009
c. Post 2009
   i. **Sale Price ($)**
      Min: 1
      1st Quartile (Q1): 635,000
      Median (Q2): 820,000
      3rd Quartile (Q3): 995,000
      Max: 2,460,000
   ii. **Gross Square Feet**
      Min: 504
      Q1: 1,228
      Median: 1,632
      Q3: 2,268
      Max: 4,916
d. Pre-2009 (before 2009)
   i. **Sale Price ($)**
      Min: 1
      Q1: 485,000
      Median: 618,000
      Q3: 750,000
      Max: 2,500,000
   ii. Gross Square Feet
      Min: 730
      Q1: 1,296
      Median: 1,775.5

Q3: 2,432
Max: 4,916

Sale Price

- Median prices are higher post 2009, $820,000 (post), vs. $618,000 (pre)
- The upper range (Q3) is also higher post 2009 ($995,000) compared to pre 2009 ($750,000), suggesting stronger top-end sales in the most recent period.
- The IQR for post 2009 is $360,000 compared to that of pre 2009 that was only $265,00. This shows a greater spread of the middle 50% after 2009. This suggests a broader range of properties being sold in the mid-market.
- The maximum price is very similar for both due to the filtering constraints.

Gross Square Feet

- The median value was higher pre 2009 (1,775.5sqft vs. 1,632sqft) than it was post 2009.
- The IQR was greater pre 2009 (1,136sqft vs 1,040 sqft) than post 2009, meaning there was more variability in the size of properties sold in the middle 50% of listings.
- The maximum square footage remains the same due to the filtering constraints.

e. Standard deviation of prices
   Post 2009: $354,762.10
   Pre 2009: $250,536

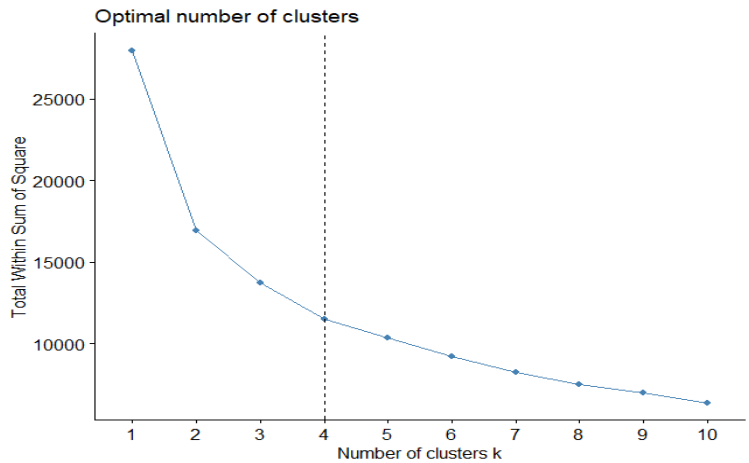
The post 2009 sale price has a higher standard deviation, showing that prices became more variable after 2009. This suggests a wider range or property sales, reflecting a wider mix of properties or market conditions.

f. Correlation Values
   Post 2009 Correlation: 0.4833041
   Pre 2009 Correlation: 0.4774604

The correlation coefficient remains very similar before and after 2009. The correlation indicates a moderate relationship between the price and the gross square footage. Suggesting that as the gross square footage increases, the sale price will also increase. This shows that whilst it has some positive correlation other factors may also impact price (location, amenities, building condition etc)

Question 2.

Graph showing optimal amount of clusters = 4



The 4 clusters were made using the property sale price, gross square footage and age when sold. I removed both residential units from the means as they were hindering the ability to create meaningful clusters as the data was not sufficiently increasing insights. This is likely due to the fact that only some properties have units.

Below is the table with the scale cluster values, assigned to the bayside data. They represent the cluster groups that were used for the graphs.

	SALE_PRICE	GROSS_SQUARE_FEET	PROPERTY_AGE
1	-0.76662889	-0.6649312	0.06649714
2	0.04075069	1.1367830	-1.33786276
3	1.93048595	1.4281902	-0.37753639
4	0.28816447	-0.2572925	0.73647380

Cluster 1 contains properties that are cheaper and smaller than the average house in the area. Their average property age is comparable to average for others in the area. The houses are likely smaller, and more budget for houses in the area. Cluster 2 contains properties that are much larger and newer than the average properties in the area. Their average price is slightly below average than others in the area. The properties are larger, good value for size houses. Cluster 3 contains properties that are much larger than the other properties in the area. Their age on average is slightly below the average for others in the area. The properties are larger, more expensive properties in the area, and are likely seen as luxury properties for the area. Cluster 4 contains properties that are older than others in the area on average. Their prices and property sizes are slightly smaller than other properties in the area. The properties are likely older, potentially differentiated in style due to their age.

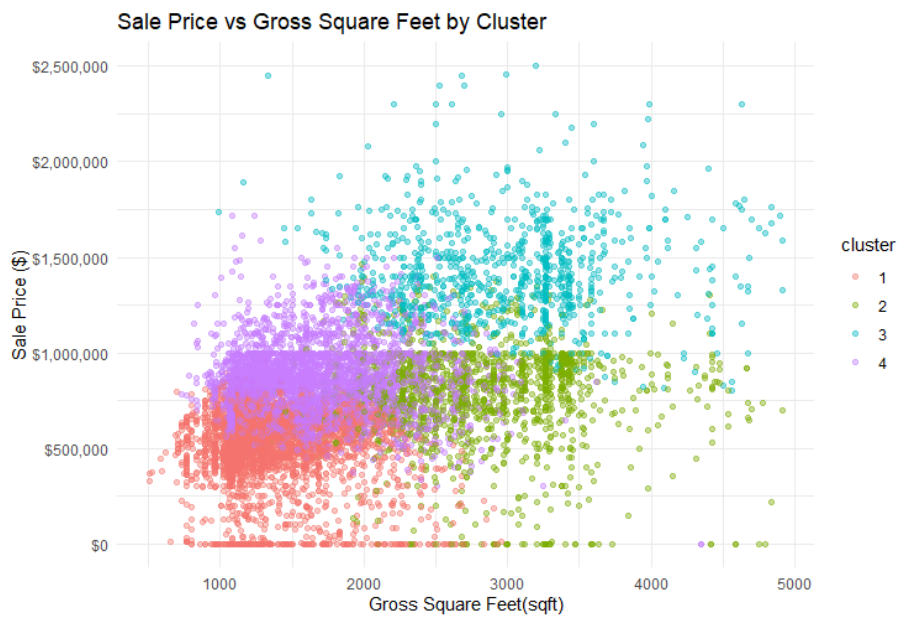
Cluster 1 - Budget Compacts

Cluster 2 - Smart Spacers

Cluster 3 - Modern Luxury

Cluster 4 - Vintage Charmers

Graph 1.

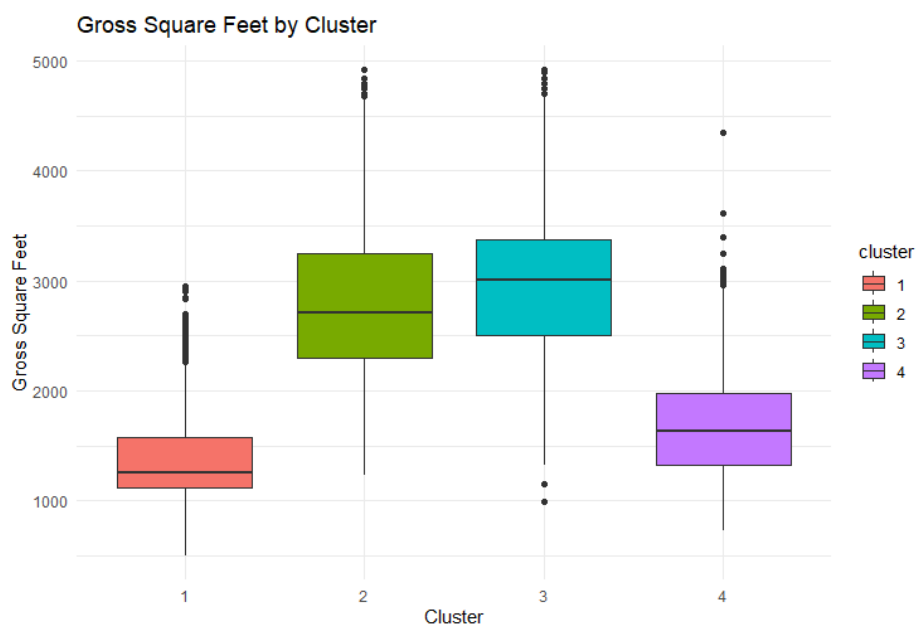


This plot shows the distribution of each of the 4 clusters, based on the gross square footage and the sale price. It gives an indication of their distribution between the two variables, per cluster. It gives an indication of sale price, gross square footage and cluster. It helps gain a visual understanding of the clustering at work.

Graph 2.

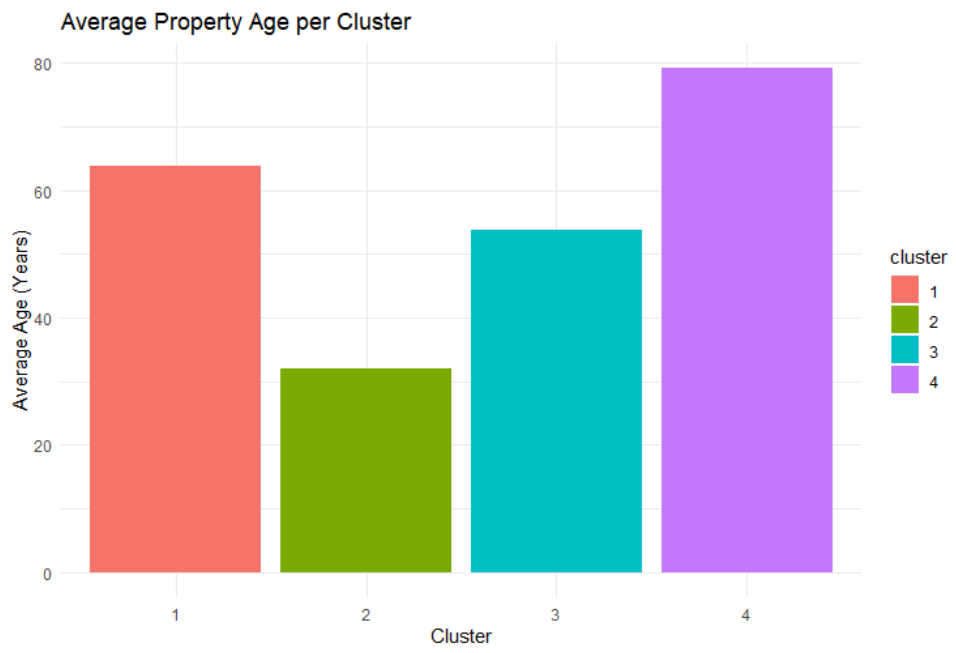
This plot shows the distribution of each of the 4 clusters, based on the property age at sale and the sale price. It gives an indication of their distribution between the two variables, per cluster. It gives an indication of the relationship between age, price and cluster. It helps gain a visual understanding of the clustering at work.

Graph 3.



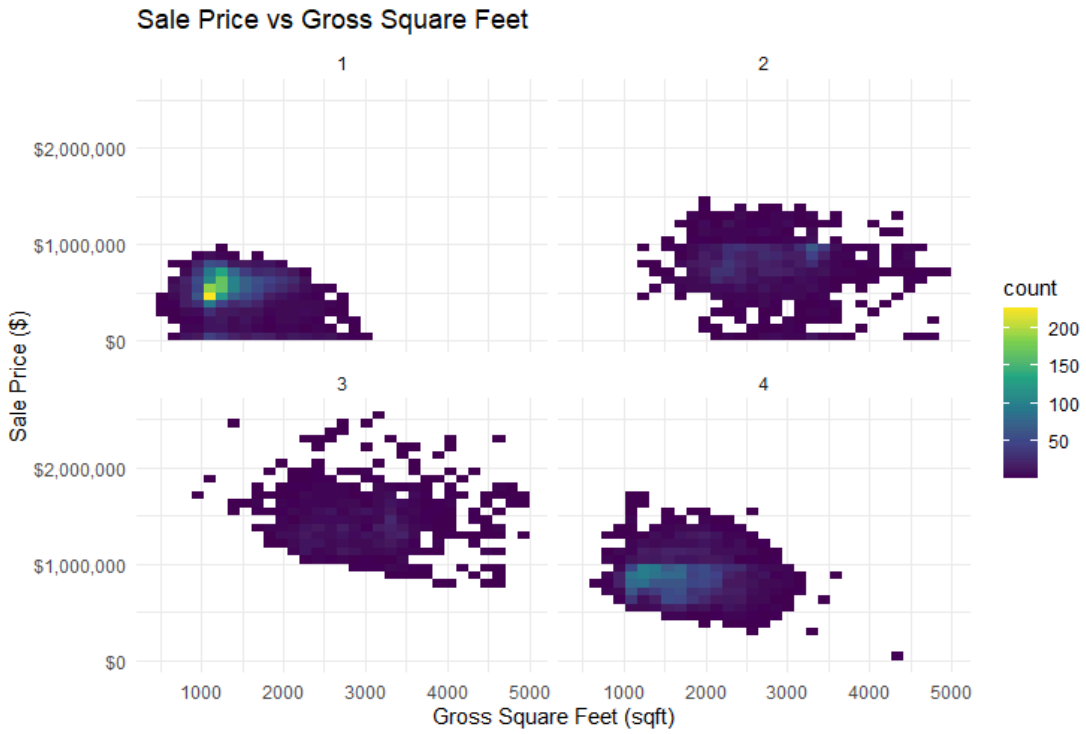
This box and whisker plot shows the distribution of the gross square footage in relation to the min, Q1, median, Q3 and maximum values. It shows how the gross square footage impacts the groups for clustering, with clear differences of each cluster.

Graph 4.



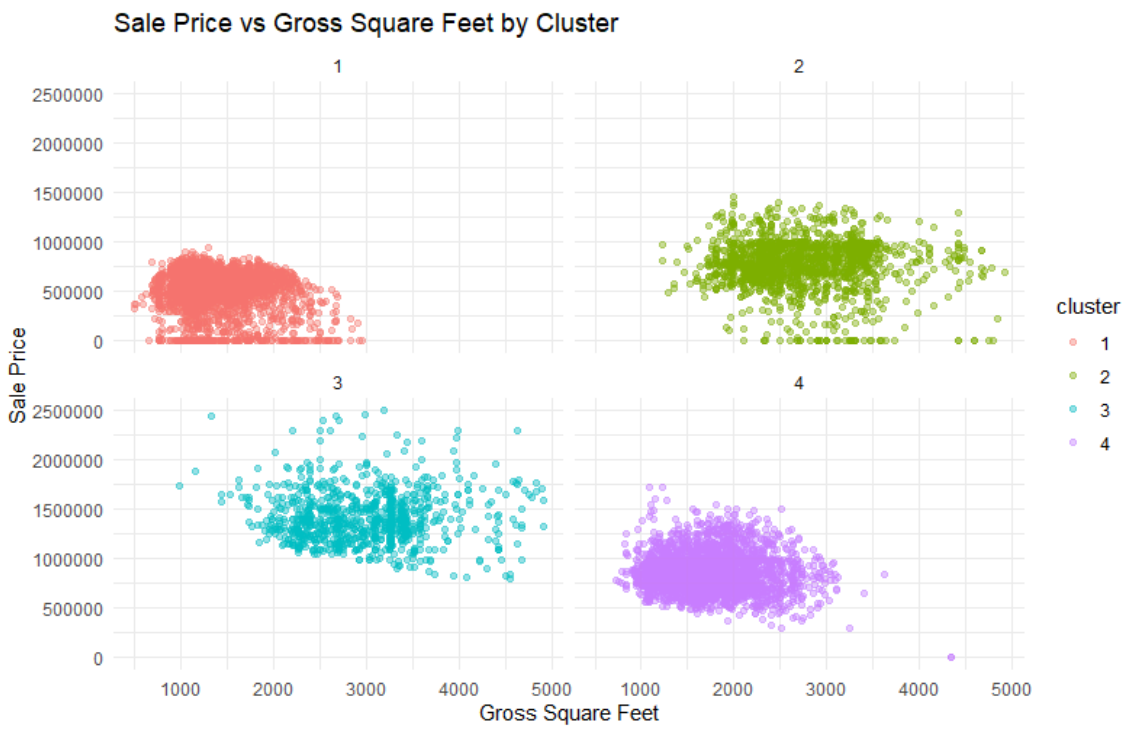
This bar chart shows the average age per cluster, indicating the relationship between how the clusters were grouped together based on the impact of the property age variable. It demonstrates a clear difference between each group.

Graph 5. (Bonus)



This bonus graph shows the clusters broken down by each grouping in a scatter plot, using bin2d as a different visual representation. It again uses price and gross square footage.

Graph 6. (Bonus)



This bonus graph shows the facet scatter graph, showing the clusters broken down by price and gross square footage. This is the same repeated data as above, but in a different visual format to demonstrate the different ways of displaying insights.

Question 3.

The most common building classes included:

- A1 - One-Family Dwellings
- A5 - Two Family Dwellings

T-test on Sale Price

- Mean sale price A1: $536,109
- Mean sale price A5: $389,504
- $t = 14.91$, $p < 2.2e-16$, meaning it is statistically significant
- 95% CI for difference: $127,335 to $165,875

One-family dwellings (A1) sell for significantly more than two-family dwellings (A5), with an average price difference between $127k and $166k. This difference is highly statistically significant.

T-test on Gross Square Feet

- Mean size A1: 1,937 sq ft
- Mean size A5: 1,262 sq ft
- $t = 52.57$, $p < 2.2e-16$, meaning it is statistically significant
- 95% CI for difference: 650 to 700 sq ft

One-family dwellings (A1) are significantly larger on average than two-family dwellings (A5), with an average size difference of about 650–700 sq ft. Again, this difference is highly statistically significant.

In summary, there are meaningful, statistically significant differences between A1 and A5 dwellings in both sale price and square footage, with A1 dwellings being both more expensive and larger.

CODE:

```
library(tidyverse)
library(dplyr)
library(ggplot2)
library(scales)
library(lubridate)
library(factoextra)
library(patchwork)

BOROUGH <- read_csv("BOROUGH.csv")
BUILDING_CLASS <- read_csv("BUILDING_CLASS.csv")
NEIGHBORHOOD <-read_csv("NEIGHBORHOOD.csv")
NYC_TRANSACTION_DATA <- read_csv("NYC_TRANSACTION_DATA.csv")

all_nyc <- NYC_TRANSACTION_DATA %>% left_join(BUILDING_CLASS, by
=c("BUILDING_CLASS_FINAL_ROLL" =
                                                    "BUILDING_CODE_ID")) %>%
left_join(NEIGHBORHOOD, by=c("NEIGHBORHOOD_ID" = "NEIGHBORHOOD_ID"))
bayside <- filter(all_nyc, NEIGHBORHOOD_NAME == "BAYSIDE")

## 1. DESCRIPTIVE ANALYSIS
bayside_residential <- bayside%>%
  filter(TYPE == "RESIDENTIAL",
        SALE_PRICE > 0,
        SALE_PRICE <= 2500000,
        GROSS_SQUARE_FEET > 0,
        GROSS_SQUARE_FEET <= 5000,
        )

bayside_residential$SALE_DATE <- as.Date(bayside_residential$SALE_DATE, format =
"%Y-%m-%d")

bayside_res_post_2009 <- bayside_residential %>%
  filter(bayside_residential$SALE_DATE >= as.Date("2009-01-01"))

bayside_res_pre_2009 <- bayside_residential %>%
  filter(bayside_residential$SALE_DATE < as.Date("2009-01-01"))

nrow(bayside_res_post_2009)

mean(bayside_res_post_2009$SALE_PRICE, na.rm = TRUE)
mean(bayside_res_post_2009$GROSS_SQUARE_FEET, na.rm = TRUE)

mean(bayside_res_pre_2009$SALE_PRICE, na.rm = TRUE)
```

```
mean(bayside_res_pre_2009$GROSS_SQUARE_FEET, na.rm = TRUE)

mean(bayside_res_post_2009$SALE_PRICE, na.rm = TRUE) -
mean(bayside_res_pre_2009$SALE_PRICE, na.rm = TRUE)
mean(bayside_res_post_2009$GROSS_SQUARE_FEET, na.rm = TRUE) -
mean(bayside_res_pre_2009$GROSS_SQUARE_FEET, na.rm = TRUE)

fivenum(bayside_res_post_2009$SALE_PRICE, na.rm = TRUE)
fivenum(bayside_res_post_2009$GROSS_SQUARE_FEET, na.rm = TRUE)

fivenum(bayside_res_pre_2009$SALE_PRICE, na.rm = TRUE)
fivenum(bayside_res_pre_2009$GROSS_SQUARE_FEET, na.rm = TRUE)

sd(bayside_res_post_2009$SALE_PRICE, na.rm = TRUE)
sd(bayside_res_pre_2009$SALE_PRICE, na.rm = TRUE)

cor(bayside_res_post_2009$SALE_PRICE, bayside_res_post_2009$GROSS_SQUARE_FEET,
use = "complete.obs")
cor(bayside_res_pre_2009$SALE_PRICE,
bayside_res_pre_2009$GROSS_SQUARE_FEET,use = "complete.obs" )

## 2. KMEANS

bayside_kmeans <- bayside %>%
  filter(TYPE == "RESIDENTIAL",
         SALE_PRICE > 0,
         SALE_PRICE <= 2500000,
         GROSS_SQUARE_FEET > 0,
         GROSS_SQUARE_FEET < 5000,
         YEAR_BUILT > 0) %>%
  mutate(
    SALE_YEAR = as.numeric(format(as.Date(SALE_DATE), "%Y")),
    SALE_PRICE = as.numeric(SALE_PRICE),
    GROSS_SQUARE_FEET = as.numeric(GROSS_SQUARE_FEET),
    PROPERTY_AGE = SALE_YEAR - as.numeric(YEAR_BUILT)
  ) %>%
  select(SALE_PRICE, GROSS_SQUARE_FEET, PROPERTY_AGE) %>%
  na.omit()

scaled_bayside_residential <- scale(bayside_kmeans)
df_scaled_bayside_residential <- as.data.frame(scaled_bayside_residential)

fviz_nbclust(df_scaled_bayside_residential, kmeans, method = "wss") +
  geom_vline(xintercept = 4, linetype = 2)
```

```
set.seed(123)
model_1 <- kmeans(df_scaled_bayside_residential, centers = 4,nstart = 100)
print(model_1$centers)
bayside_kmeans$cluster <- as.factor(model_1$cluster)

# GRAPH 1
ggplot(bayside_kmeans, aes(x = GROSS_SQUARE_FEET, y = SALE_PRICE, color = cluster))
+
  geom_point(alpha = 0.4) +
  scale_y_continuous(labels = dollar) +
  labs(title = "Sale Price vs Gross Square Feet by Cluster",
       x = "Gross Square Feet(sqft)", y = "Sale Price ($)") +
        theme_minimal()
# GRAPH 2
ggplot(bayside_kmeans, aes(x = PROPERTY_AGE, y = SALE_PRICE, color = cluster)) +
  geom_point(alpha = 0.4) +
  labs(title = "Sale Price vs Age by Cluster",
       x = "Age (Years)", y = "Sale Price ($)") +
  theme_minimal()

# GRAPH 3
ggplot(bayside_kmeans, aes(x = cluster, y = GROSS_SQUARE_FEET, fill = cluster)) +
  geom_boxplot() +
  labs(title = "Gross Square Feet by Cluster", x = "Cluster", y = "Gross Square Feet") +
  theme_minimal()

# GRAPH 4
bayside_kmeans %>%
  group_by(cluster) %>%
  summarise(avg_age = mean(PROPERTY_AGE)) %>%
  ggplot(aes(x = cluster, y = avg_age, fill = cluster)) +
  geom_col() +
  labs(title = "Average Property Age per Cluster", x = "Cluster", y = "Average Age (Years)") +
  theme_minimal()

# BONUS GRAPH 5
ggplot(bayside_kmeans, aes(x = GROSS_SQUARE_FEET, y = SALE_PRICE)) +
  geom_bin2d() +
  scale_fill_viridis_c() +
  scale_y_continuous(labels = dollar) +
  labs(title = "Sale Price vs Gross Square Feet",
       x = "Gross Square Feet (sqft)", y = "Sale Price ($)") +
  facet_wrap(~ cluster) +
```

```
  theme_minimal()

# BONUS GRAPH 6
facet_scatter <- ggplot(bayside_kmeans, aes(x = GROSS_SQUARE_FEET, y = SALE_PRICE,
color = cluster)) +
  geom_point(alpha = 0.4) +
  facet_wrap(~ cluster) +
  labs(title = "Sale Price vs Gross Square Feet by Cluster",
       x = "Gross Square Feet", y = "Sale Price") +
  theme_minimal()
facet_scatter

## 3. T-TEST
topclasses <- bayside %>%
  filter(TYPE == "RESIDENTIAL",
         SALE_PRICE > 0,
         SALE_PRICE <= 2500000,
         GROSS_SQUARE_FEET > 0,
         GROSS_SQUARE_FEET < 5000,
         ) %>%
  count(BUILDING_CLASS_FINAL_ROLL, sort = TRUE) %>%
  slice(1:2) %>%
  pull(BUILDING_CLASS_FINAL_ROLL)


top2_filtered <- bayside %>%
  filter(TYPE == "RESIDENTIAL", BUILDING_CLASS_FINAL_ROLL %in% topclasses)

table(top2_filtered$BUILDING_CLASS_FINAL_ROLL)

t_test_saleprice <- t.test(
  SALE_PRICE ~ BUILDING_CLASS_FINAL_ROLL,
  data = top2_filtered
)

print(t_test_saleprice)

t_test_sqft <- t.test(
  GROSS_SQUARE_FEET ~ BUILDING_CLASS_FINAL_ROLL,
  data = top2_filtered
)

print(t_test_sqft)
```